Iman Gholami

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Research Interests

Approximation Algorithms, Network Design, Combinatorial Optimization, Algorithmic Game Theory, Metric Distortion, Formal Verification, AI for Mathematics

Education

University of Maryland, College Park — MD, USA
PhD Candidate in Computer Science — *Jan 2023 – Present*
MSc in Computer Science — *May 2025*
- GPA: 3.97

Sharif University of Technology — Tehran, Iran
BSc in Computer Engineering — *May 2022*
- GPA: 3.57

Experience

Research Assistant — Jan 2023 – Present
University of Maryland — *College Park, MD*
- Research on network design problems, including variations of classic connectivity problems, for better approximation factors than state-of-the-art algorithms.
- DARPA expMath team research towards verifiable mathematical proving through Artificial Intelligence (AI).
- Research on distortion problems, specifically classic problems with access to ordinal information instead of cardinal information.
- DARPA QuICC team research on Quantum-Inspired solvers for complex optimization problems.
- Academic mentoring of incoming PhD students and coordination of research teamwork.
- External reviewer for major conferences including FOCS, STOC, SODA, SPAA, and CDC.

Teaching Assistant — Jan 2023 – Present
University of Maryland, College Park — *MD, USA*
- Computational Genomics (Rob Patro, Spring 2026)
- Introduction to Computational Game Theory (MohammadTaghi Hajiaghayi, Fall 2025)
- Computer and Network Security (Michael Marsh, Fall 2024)
- Introduction to Computational Game Theory (MohammadTaghi Hajiaghayi, Fall 2023)
- Algorithms (Max Morowski & Clyde Kruskal, Spring 2023)

Software Engineer Intern — Jun 2026 – Aug 2026
Intrinsic Technologies (Axiom) — *New York, NY*
- Advancing formal verification of Zero-Knowledge Virtual Machines (ZK VMs) in Lean 4.
- Designed and implemented a Reinforcement Learning (RL) evaluation system to optimize Large Language Models (LLMs) and reasoning models for automated proof generation.

Software Engineer Intern — May 2025 – Aug 2025
Axal (YC W25) — *San Francisco, CA*
- Design and implement code analysis tools for information extraction using Artificial Intelligence (AI), with the goal of rewriting legacy code into modern, maintainable systems.

Software Engineer and Team Leader — Nov 2018 – Dec 2022
Balad (Local Navigation and Maps App, 30M+ Installs) — *Tehran, Iran*

Teaching Assistant — Sep 2019 – Jun 2022
Sharif University of Technology — *Tehran, Iran*
- Theory of Machines and Languages (Ali Movaghar, Fall 2021)
- Design Algorithm (Hamid Zarrabi-Zadeh, Fall 2021)
- Modern Information Retrieval (Mahdieh Soleymani Baghshah, Spring 2021)
- Discrete Structures (Hamid Zarrabi-Zadeh, Spring 2021)
- Design Algorithm (Mohammad Ali Abam, Fall 2019)

Software Engineer Intern — Jul 2018 – Oct 2018
Bazaar (Local Application Marketplace, 100M+ Installs) — *Tehran, Iran*

Honors/Awards

Best Paper Award FOCS 2025
Competitive University Fellowships (UMD Dean's Fellowship, MPower Early Scholars Award)
Silver Medal in International Olympiad in Informatics (IOI 2017)
2nd Place in ICPC NAS, Mid-Atlantic USA Region 2023
2nd Place in ACM/ICPC Asia Region, Tehran Site 2017
4th Place in ACM/ICPC Asia Region, Tehran Site 2020
Gold Medal in Iran National Olympiad in Informatics (INOI 2016)
Silver Medal in Iran National Olympiad in Informatics (INOI 2015)
Bronze Medal in National Olympiad of Computer Engineering 2021

Publications

Nonuniform Buy-at-Bulk Network Design: Improved Approximation and LP Integrality Gap
arXiv preprint, 2026 | https://github.com/igholami/buy-at-bulk-lean

The Tight Competitive Ratio of Greedy for Online Steiner Forest
arXiv preprint, 2026 | https://github.com/igholami/online-steiner-forest-lean

Constructive Arboricity in Almost-Linear Time for Dense Graphs
arXiv preprint, 2026

Beyond the Library: An Agentic Framework for Autoformalizing Research Mathematics
arXiv preprint, 2026

Quiet Planting for k-SAT, Multiple Solutions of Arbitrary Geometry
Conference on Learning Theory (COLT 2026)

Bi-Criteria Metric Distortion
International Conference on Learning Representations (ICLR 2026)

Breaking a Long-Standing Barrier: 2-ε Approximation for Steiner Forest
IEEE Symposium on Foundations of Computer Science (FOCS 2025)

2-Approximation for Prize-Collecting Steiner Forest
Journal of the ACM (JACM 2025), ACM-SIAM Symposium on Discrete Algorithms (SODA 2024)

Prize-Collecting Forest with Submodular Penalties: Improved Approximation
Integer Programming and Combinatorial Optimization (IPCO 2025)

Prize-Collecting Steiner Tree: A 1.79 Approximation
ACM Symposium on Theory of Computing (STOC 2024)

Skills

Research & Leadership: Technical Leadership, Teamwork, Communication, Conflict Resolution, Academic Mentoring, Research Strategy, Cross-functional Collaboration

Theoretical Computer Science (TCS): Approximation Algorithms, Network Design, Algorithmic Game Theory, Combinatorial Optimization, Graph Theory, Data Structures, Problem Solving

AI & Machine Learning: Large Language Models (LLMs), Automated Theorem Proving, Generative AI, AI-driven Analysis, Applied ML

Software Engineering: Distributed Systems, System Architecture, Static Analysis, Large-Scale Backend Systems, High-Performance Computing, API Design

Languages: C++ (200K+ LOC), Python (200K+ LOC), Lean 4, Kotlin, TypeScript, Go, Java, Bash, SQL

Frameworks & Libraries: React, FastAPI, Django, Node.js, NumPy, Pandas, Scikit-learn

Cloud, DB & Tools: AWS, Docker, Kubernetes, PostgreSQL, MongoDB, Redis, Git, CI/CD, Linux, Nginx

References

Prof. Hajiaghayi, MohammadTaghi hajiagha@umd.edu
Minker Professor at Computer Science Department, University of Maryland, College Park

Dr. Chakraborty, Diptarka diptarka@comp.nus.edu.sg
Assistant Professor at School of Computing, National University of Singapore

Dr. Abam, Mohammad Ali abam@sharif.edu
Associate Professor at Computer Engineering Department, Sharif University of Technology

Dr. Zarrabi-Zadeh, Hamid zarrabi@sharif.edu
Associate Professor at Computer Engineering Department, Sharif University of Technology